

VAANI: Capturing the language landscape for an inclusive digital India

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Abstract

Project VAANI is an initiative to create an India-representative multi-modal dataset that comprehensively maps India’s linguistic diversity, starting with 165 districts across the country in its first two phases. Speech data is collected through a carefully structured process that uses image-based prompts to encourage spontaneous responses. Images are captured through a separate process that encompasses a broad range of topics, gathered from both within and across districts. The collected data undergoes a rigorous multi-stage quality evaluation, including both automated and manual checks to ensure highest possible standards in audio quality and transcription accuracy. Following this thorough validation, we have open-sourced around 289K images, approximately 31,270 hours of audio recordings, and around 2,067 hours of transcribed speech, encompassing 112 languages from 165 districts from 31 States and Union territories. Notably, significant of these languages are being represented for the first time in a dataset of this scale, making the VAANI project a groundbreaking effort in preserving and promoting linguistic inclusivity. This data can be instrumental in building inclusive speech models for India, and in advancing research and development

across speech, image, and multimodal applications.

1 Introduction

India is one of the most linguistically diverse countries in the world. This linguistic diversity arises from a range of factors including historical and cultural influences, geographical dispersion, ethnic and social diversity, political-administrative needs and religious variations (Kidwai, 2019). Nearly 1.4 billion people in India communicate using a range of languages and dialects, often shaped by their regional and cultural context. Multilingualism is the norm, with many Indians proficient in more than one language. According to the 2011 census (of India, 2011), after detailed linguistic scrutiny and rationalization, 1,369 mother tongues were identified. After linguistic scrutiny, edit and rationalization, and those spoken by over 10,000 individuals were grouped into 121 languages. These belong to five language families: Indo-European (23 languages, 78.05% of the population), Dravidian (17 languages, 19.64%), Austro-Asiatic (14 languages, 1.11%), Tibeto-Burmese (66 languages, 1.01%), and Semito-Hamitic (1 language, <0.01%). Among these, 22 languages are officially recognized under the

Eighth Schedule of the Indian Constitution (of India, n.d.).

In multilingual contexts, voice-based technologies play a critical role in bridging accessibility gaps, particularly in domains such as education, governance, and health care (Kumar and Agarwal, 2012)(Brewer et al., 2005). Speech technologies enable interaction between diverse social groups who speak different languages. By reducing dependence on text-based literacy and supporting communication in native languages, speech interfaces promote greater inclusion in digital spaces (Wu et al., 2025).

Recent advancements in Artificial Intelligence (AI), particularly in generative AI, have revolutionized how we process and generate language. Techniques like attention mechanisms(Vaswani et al., 2017) and diffusion models (Ho et al., 2020), coupled with self-supervised pretraining followed by fine-tuning, have driven significant progress (Lee et al., 2023). However, these models rely heavily on high-quality and representative training data.

Language modeling efforts often treat “language” as a singular, monolithic label (Backus, 1999). In practice, spoken language varies significantly across regions, communities, education levels, and genders, even within the same official language (Shapiro and Schiffman, 2008)(Sailaja, 2012). Thus, datasets anchored only on language identifiers miss important regional and sociolinguistic variations. To develop inclusive and robust models, we need data that reflects both language-anchored and region-anchored variations in speech patterns.

2 Related works

Data collection for speech technology typically follows two main strategies: read speech and spontaneous speech. In read speech, speakers recite predetermined text, which simplifies organization and annotation, since the content is controlled and consistent. This approach is particularly use-

ful for creating initial datasets where linguistic clarity is key. In contrast, spontaneous speech captures natural, unscripted responses from speakers based on stimuli, offering richer linguistic diversity and more authentic language usage. However, spontaneous speech presents greater challenges, as it requires extensive transcription and annotation, making the process more resource-intensive. This method is crucial for developing datasets that reflect real-world communication but demands substantial effort to meet specific use case needs.

Thanks to the efforts of research communities and various organizations, many datasets are now available for Indic languages, each with varying degrees of coverage. Some notable datasets include: IndicVoice (Javed et al., 2024), Shrutilipi (Bhogale et al., 2023), Kathbath (Javed et al., 2023), Spring-inx (Gangwar et al., 2023), FLEURS (Conneau et al., 2023), CommonVoice(Ardila et al., 2019), IndicTTS(Baby et al., 2016), CMU Wilderness Multilingual Speech Dataset (Black, 2019) and MUCS Dataset(Diwan et al., 2021a). Although there are multiple datasets, gaps remain in ensuring their inclusion in languages and communities. Some key drawbacks are outlined below.

Lack of extensive language coverage: Efforts to collect speech data for Indic languages have predominantly focused on major languages, mainly those recognized in the Indian constitution. These efforts often concentrate on the 22 officially recognized languages, which limits inclusivity and representation of the country’s vast linguistic diversity.

Lack of geographic and other demographic coverage: Language is a continuum — it varies significantly based on geography. Within a single language, we often observe substantial variation from place to place. Most existing datasets do not adequately account for this diversity and have failed to capture the regional and dialectal differences present within languages. Given the high levels of linguistic and geographical diversity

Dataset	Year	Selection Method	Duration (Hours)	No. Langs	Languages Covered
IndicVoice	2024	Mixed	19550	22	as, bn, bho, gu, hi, kn, kok, ml, mr, or, pa, raj, ta, te, ur, etc.
Shrutilipi	2023	Read	6457	12	as, bn, gu, hi, kn, ml, mr, or, pa, ta, te, ur
Kathbath	2023	Read	1684	12	bn, gu, hi, kn, ml, mr, or, pa, ta, te, ur, as
Spring-inx	2023	Mixed	2005	10	bn, hi, kn, ml, mr, or, pa, ta, te, ur
FLEURS	2023	Mixed	163	13	as, bn, gu, hi, kn, ml, mr, or, pa, sd, ta, te, ur
CommonVoice	2023	Read	373	10	as, bn, gu, hi, kn, ml, mr, or, ta, te
CMU Wilder-ness	2023	Read	286	18	as, bn, gu, hi, kn, ml, mr, or, pa, ta, te, ur, etc.
MUCS Dataset	2023	Mixed	8	8	as, bn, gu, hi, kn, mr, ta, te
IndicTTS	2023	Read	225	13	as, bn, gu, hi, kn, ml, mni, mr, or, raj, ta, te

Table 1: Indic Corpus. Language codes: as (Assamese), bn (Bengali), bho (Bhojpuri), gu (Gujarati), hi (Hindi), kn (Kannada), kok (Konkani), ml (Malayalam), mr (Marathi), or (Odia), pa (Punjabi), raj (Rajasthani), ta (Tamil), te (Telugu), ur (Urdu), sd (Sindhi), mni (Manipuri).

across India, such datasets may not truly represent the wide range of speech patterns found across the country.

Lack of multimodel data: Furthermore, AI is increasingly moving toward multimodal systems, which require datasets covering different modalities. However, most publicly available speech focused datasets for Indic languages are limited to a two-modality setup, speech and text. Building multimodal datasets will be crucial to enable the development of robust multimodal models tailored for Indic languages and contexts.

To address this gap, the VAANI initiative adopts a geo-centric data collection approach to build a more inclusive and representative dataset. This method is specifically designed to capture the rich linguistic diversity of India, including low-resource and underrepresented languages. By prioritizing inclusivity and diversity, VAANI contributes to the development of equitable AI so-

lutions for language technologies. Its innovative prompting strategy enables the creation of a multimodal dataset comprising images, speech, and text, which lays the groundwork for building robust multimodal models tailored to Indic languages.

3 The Dataset

VAANI is an India-representative multimodal, multilingual dataset. It contains audio recordings collected along with the images shown to the speakers, and where available, the corresponding transcriptions. Each speaker was presented with an image and asked to describe it in their own words, enabling the creation of aligned image–speech–text triplets.

Phase 1 and Phase 2 of the dataset include data collected from 165 districts across the country, covering 112 languages. It comprises 24,009,427 audio segments collected from 158,441 speakers

recorded in response to 289,838 images, with a total audio duration of approximately 31,270 hours. From this collection, 2,067 hours of audio have been manually transcribed, with transcription data distributed nearly evenly across the 165 districts, ensuring balanced geographic representation.

3.1 District represented

The data has been collected from 165 districts across 28 states and 3 union territories in India.

In southern India, Andhra Pradesh includes Anantpur, Annamaya, Chittoor, Guntur, Krishna, Manyam, Srikakulam, Sri Satya Sai, Vishakhapatnam, and Adilabad. Karnataka contributes Bangalore, Belgaum, Bellary, Bijapur, Bidar, Chamrajnagar, Dakshina Kannada, Dharwad, Gulbarga, Koppal, Mysore, Raichur, and Shimoga. Kerala is represented by Kasaragod, Kollam, Kozhikode, Thiruvananthapuram, and Wayanad. Goa is represented by North Goa and South Goa. Tamil Nadu contributes Chennai, Kanyakumari, Namakkal, and Nilgiris. Telangana includes Hyderabad, Karimnagar, Komaram Bheem, Mahabubabad, and Nalgonda.

In the North-East, Arunachal Pradesh contributes Longding, Lower Dibang Valley, and Papum Pare. Assam includes Hailakandi, Kamrup Metropolitan, Karbi Anglong, and Sonitpur. Meghalaya includes North Garo Hills, South Garo Hills, and West Garo Hills. Nagaland contributes Dimapur and Kohima. Tripura includes Dhalai, South Tripura, Unakoti, and West Tripura. Mizoram is represented by Aizawl, and Sikkim by Gangtok. Manipur includes Imphal West.

Eastern India includes Bihar with Araria, Begusarai, Bhagalpur, Darbhanga, East Champaran, Gaya, Gopalganj, Jahanabad, Jamui, Kaimur, Katihar, Kishanganj, Lakhisarai, Madhepura, Muzaffarpur, Patna, Purnia, Saharsa, Samastipur, Saran, Sitamarhi, Supaul, Vaishali, and West Champaran. Jharkhand is represented by Deoghar, Garhwa, Jamtara, Palamu, Ranchi, and Sahebganj. Odisha includes Khordha, Malkangiri, Nuapada,

and Sambalpur. West Bengal includes Alipurduar, Cooch Behar, Darjeeling, Dakshin Dinajpur, Jalpaiguri, Jhargram, Kolkata, Malda, North 24 Parganas, Paschim Medinipur, and Purulia.

In the West, Gujarat includes Devbhoomi Dwarka, Gandhinagar, Navsari, and Valsad. Rajasthan includes Barmer, Bikaner, Churu, Jaipur, Jaisalmer, and Nagaur. Maharashtra includes Aurangabad, Chandrapur, Dhule, Gondia, Mumbai Suburban, Nagpur, Sindhudurg, Pune, Solapur, and Washim.

In northern India, Haryana includes Charkhi Dadri, Jhajjar, and Rohtak, while Himachal Pradesh contributes Shimla. Punjab includes Fazilka, Kapurthala, and Pathankot. Uttarakhand contributes Tehri Garhwal and Uttarkashi. Uttar Pradesh includes Budaun, Deoria, Etah, Ghazipur, Gorakhpur, Hamirpur, Jalaun, Jyotiba Phule Nagar, Lalitpur, Lucknow, Muzaffarnagar, Saharanpur, Shamli, and Varanasi.

In central India, Madhya Pradesh includes Bhopal, Dhar, Katni, and Umari. Chhattisgarh includes Balrampur, Bastar, Bilaspur, Jashpur, Kabirdham, Korba, Koriya, Narayanpur, Raigarh, Raipur, Rajnandgaon, Sarguja, and Sukma.

Lastly, the districts from the three UTs include Chandigarh, New Delhi, Srinagar.

The districts were selected to ensure maximum coverage of the languages reported in the 2011 Census of India (of India, 2011). For each district, approximately 200 hours of audio data were collected, with around 5-10% manually transcribed. To capture regional linguistic variations, data was gathered from multiple locations within each district.

3.2 Languages captured

The dataset covers 112 languages and dialects spoken across India, reflecting its rich linguistic diversity. These are Agariya, Angami, Angika, Ao, Assamese, Awadhi, Baghati, Bagheli, Bagri, Bajjika, Bearybashe, Bengali, Bhatari, Bhili, Bhojpuri, Bihari, Bundeli, Chakhesang, Chakma,

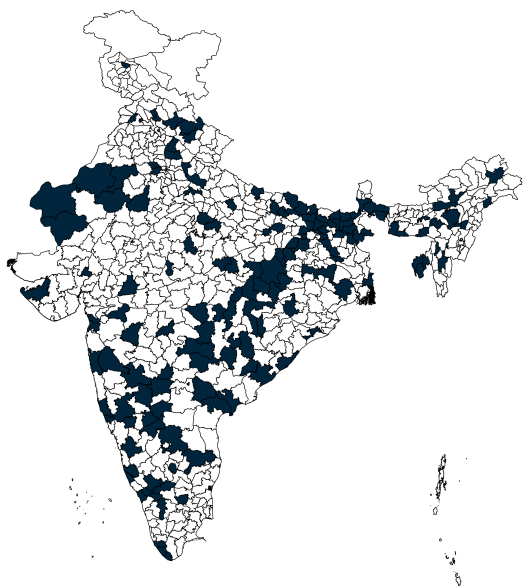


Figure 1: Districts covered in Phase 1 and Phase 2 of the Vaani Project

Chhattisgarhi, Desia, Dogari, Dogri, Dorli, Duruwa, English, Galo, Garhwali, Garo, Gondi, Gujarati, Hajong, Halbi, Harauti, Haryanvi, Hindi, Idu Mishmi, Jaipuri, Kannada, Karbi, Kashmiri, Khandeshi, Khariboli, Khortha, Kokborok, Konkani, Koya, Kuki, Kumaoni, Kurmali, Kurukh, Lambani, Lepcha, Liangmai, Liangmei, Limbu, Lotha, Magadhi, Maithili, Malayalam, Malvani, Malvi, Manipuri, Mara, Marathi, Marwadi, Meitei, Mewari, Mewati, Mizo, Nagamese, Nepali, Nimadi, Nyishi, Oriya, Pahadi, Paniya, Phom, Powari, Punjabi, Rajasthani, Rajbanshi, Rengma, Rongmei, Sadri, Sambalpuri, Sangtam, Santali, Shekhawati, Sikkimese, Sindhi, Sirmauri, Sumi, Surgujia, Surjapuri, Sylheti, Tagin, Tangkhul, Tamil, Telugu, Tenyidie, Thethi, Thadou, Tulu, Urdu, Vaiphei, Wagdi, Wancho, Yimchunger, and Zeme.

This comprehensive inclusion of both widely spoken and underrepresented languages ensures the dataset is well-suited for developing inclusive and regionally representative speech technologies.

Our unique data collection approach, where participants were encouraged to speak their na-

tive language from diverse geographies, enabled us to capture several lesser-known and under-documented languages, some of which may be digitally recorded for the first time at this scale. These languages include Angika, an Eastern Indo-Aryan language from Bihar and Jharkhand with distinct grammar and vocabulary; Kortha, a Magahi dialect spoken in the Santhal Pargana division in Jharkhand; Malvani, a dialect of Konkani recognized by many speakers as a distinct language due to its cultural and linguistic uniqueness; Shekhawati, a Rajasthani dialect spoken in Jhunjhunu, Sikar, and Churu with similarities to Marwari; Duruwa, a Central Dravidian language spoken in Odisha and Chhattisgarh; Jaipuri, another Rajasthani dialect with unique phonetic features centered in Jaipur; Bearybashe, a language of the Beary community in coastal Karnataka that blends Kannada, Tulu, and Urdu; Kurumuli, a Dravidian language of the Kuruma community in Karnataka and Kerala; Kudukh, a Munda language spoken by the Oraon tribe in Jharkhand; Bajjika, a Bhojpuri dialect with unique phonetics spoken in Vaishali and Muzaffarpur districts of Bihar; Agariya, a Hindi dialect spoken in Madhya Pradesh with distinct lexical traits; and Halbi, an Indo-Aryan language found in Chhattisgarh and Maharashtra with features separating it from standard Hindi. Apart from these, 34 rare languages were collected from North-Eastern India alone - Angami, Ao, Chakhesang, Chakma, Galo, Garo, Hajong, Idu mishmi, Karbi, Kokborok, Kuki, Liangmei, Lotha, Nagamese, Nyishi, Phom, Rengma, Rongmei, Sadri, Sangtam, Sumi, Tagin, Tenyidie, Wancho, Yimchunger, and Zeme. Many of these do not even have their own scripts.

We believe that open-sourcing these recordings will support inclusive technology development and preserve linguistic diversity.

3.3 Metadata

Each audio segment in the dataset is accompanied by the following metadata fields: 'State', 'Dis-

trict', 'Gender', 'Pincode', 'Asserted Language', and 'Languages Spoken'. These fields provide critical contextual information for organizing and analyzing the data. The 'State' and 'District' fields enable precise geographical localization, while 'Gender' supports demographic analysis. The 'Pincode' field adds further regional specificity. The 'Asserted Language' indicates the language the speaker claims to have used, ensuring alignment between transcription and linguistic context. 'Languages Spoken' captures the speaker's multilingual background, offering insight into language contact and influence. Together, these metadata supports effective categorization and enhances the utility of the dataset for building regionally and linguistically inclusive speech technologies.

Following a rigorous curation process, the dataset has been open-sourced via our web portal, with options to download data at either the state or district level. It is also available on Hugging Face to support broader research and development efforts.

4 Experiments

The Vaani dataset is multimodal and covers many Indian languages. To demonstrate its utility, we conducted a limited set of experiments on a subset of the data, using transcribed speech, untranscribed speech, and image data.

4.1 ASR Finetuning : Language Specific

The Vaani dataset comprises approximately 2,067 hours of transcribed speech across multiple languages. Here we demonstrate the performance improvements achievable by fine-tuning open-source models on part of the data separately for 4 languages and evaluate using multiple benchmark datasets to assess the consistency of the performance gains.

We fine-tuned the Whisper-small model (244M parameters)(Radford et al., 2022) separately for Hindi (331 hours), Kannada (80.244 hours), Telugu (69.039 hours), and Bengali (101 hours)

on a subset of the Vaani dataset, and evaluated each model independently on multiple benchmark datasets

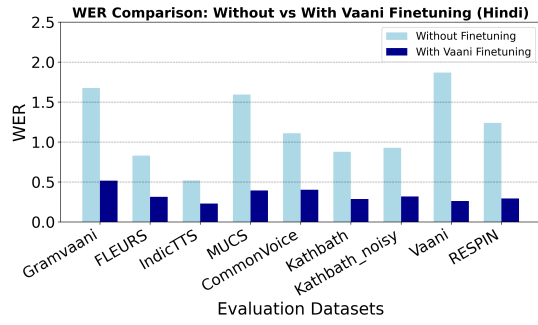
Experiments demonstrate consistent performance improvements across all languages compared to open-source models, illustrating the utility of the Vaani dataset for fine-tuning ASR models.

4.2 ASR Finetuning : Multilingual Setup

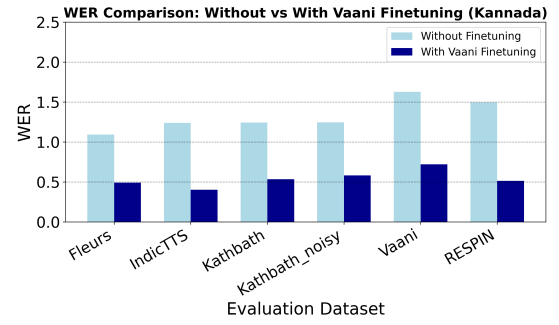
In this work, we demonstrate the performance improvements achieved through limited fine-tuning of open-source speech and multimodal models on our dataset, and we evaluate their behavior across languages to assess the consistency of these gains.

We conduct experiments using three models: Gemma-3n-2B (Team, 2025), Whisper-large-v3-turbo(Radford et al., 2022) , and parakeet-tdt-0.6b-v2(NVIDIA, 2025). All models are fine-tuned in a multilingual setting using a subset of the transcribed speech data from Hindi (593.93 hours), Kannada (120.20 hours), Telugu (113.04 hours), and Bengali (113.236 hours), along with two low-resource languages: Chakma (31.58 hours) and Bhojpuri (23.69 hours). For Gemma-3n-2B, we employ LoRA-based parameter-efficient fine-tuning, while the other models are fully fine-tuned.

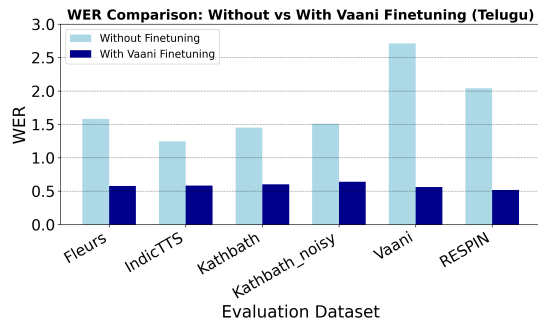
The fine-tuned models are evaluated across multiple benchmark datasets, including Vaani, to assess improvements in multilingual and low-resource speech recognition performance. Specifically, Hindi models are evaluated on GramVaani(Bhanushali et al., 2022), FLEURS, MUCS(Diwan et al., 2021b), CommonVoice, Kathbath, Kathbath-Noisy (Javed et al., 2023), VAANI, and RESPIN(Kumar et al., 2024). Bengali and Telugu models are evaluated on FLEURS, CommonVoice, Kathbath, Kathbath-Noisy, and VAANI, while Kannada models are assessed on FLEURS, Kathbath, Kathbath-Noisy, VAANI, and RESPIN. In contrast, Chakma and Bhojpuri models are evaluated solely on VAANI, due to



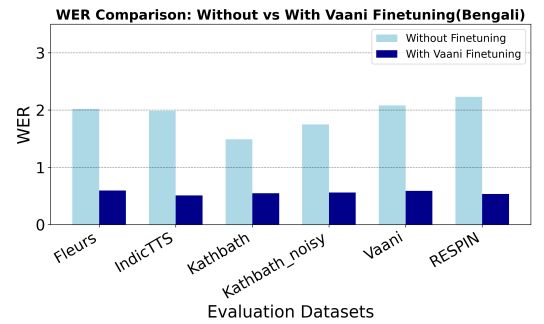
(a) Hindi



(b) Kannada



(c) Telugu



(d) Bengali

Figure 2: Word Error Rate (WER) comparison for Whisper-small models fine-tuned on the Vaani dataset across multiple languages. Each subplot shows performance on benchmark datasets for the respective language.

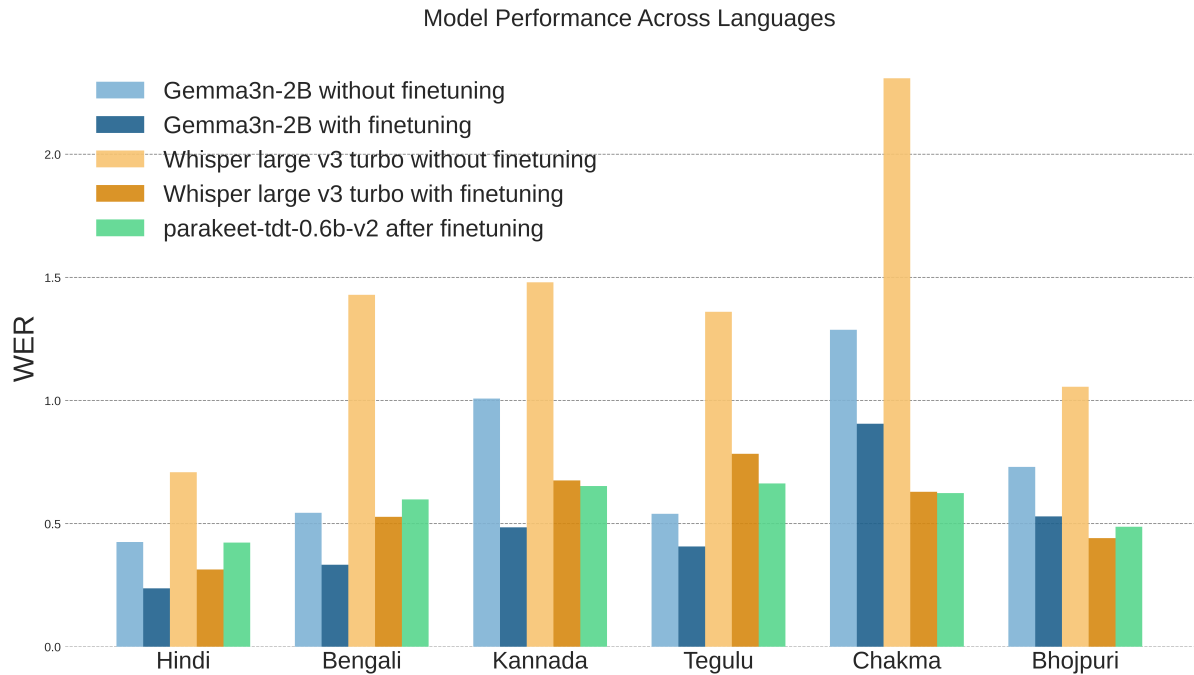


Figure 3: Average WER of models on the benchmark datasets across different languages.

the lack of other public benchmarks. We normalized the output of Gemma-3n-2B when computing WER for the non-finetuned model to remove repeated words.

The evaluation results consistently show measurable performance improvements across all models and languages after fine-tuning.

4.3 Region Specific finetuning

In order to highlight the importance of region-specific data, we fine-tuned models for the Hindi dataset at the state level, and the performance was measured across multiple states. We fine-tuned the model on Hindi data separately for each state—Bihar (130.02 hours), Uttar Pradesh (75.63 hours), Chhattisgarh (60.45 hours), Maharashtra (21.81 hours), Jharkhand (15.63 hours), Uttarakhand (10.59 hours), Rajasthan (5.54 hours), Goa (4.88 hours), and Andhra Pradesh (4.49 hours)—and evaluated performance using test data from the same states.

The evaluation results show that a model trained on data from a specific state performs comparatively better on nearby states than on distant ones, even within the same language.

4.4 Language Identification

We apply Vaani dataset to the task of spoken language identification (LID). The model architecture builds on the Whisper-medium encoder, followed by a pooled attention layer for aggregating frame-level embeddings, and a two-layer artificial neural network (ANN) classifier with a hidden size of 512. Training is performed end-to-end on our curated dataset of 42 languages, each balanced with 10 hours of speech data, using an 8:1:1 split for training, validation, and testing.

We optimize the model with AdamW at a learning rate of $1e-5$, employing linear decay with warmup, and fully fine-tune the encoder. Training is conducted for up to 2,500 steps.

The model was evaluated on the test split this curated dataset and also on two external

datasets : Kathbath(Javed et al., 2023) and FLEURS(Conneau et al., 2023). As shown in Table 2, our model achieves consistent performance across the three evaluation datasets, with overall accuracies of 74.8% (Vaani Test), 76.1% (FLEURS), and 65.8% (Kathbath). As seen in the confusion matrices in Figure 6, the model achieves high accuracy for most languages, though confusions remain among closely related dialects (e.g., Hindi–Khariboli–Maithili, Odia–Sambalpuri, and Bajjika–Magahi–Bhojpuri), highlighting the challenge of fine-grained language classification.

4.5 Image Retrieval

We aimed to explore the image modality within the Vaani dataset, which contains over 240K images collected specifically for this purpose. Since speakers were prompted to speak about images, the corresponding audio often includes details related to the visual content. Multiple speakers may describe the same image, and parts of these utterances are transcribed. For this experiment, we aggregated the transcribed segments for each image and used Large language Model(LLM) to generate a single consolidated description.

For the experiments, we used the SigLIP2 (Tschannen et al., 2025) model and the Hindi portion of the dataset. The model was fine-tuned on image–Hindi transcription pairs, using approximately 45,000 images for training, and evaluated on an image retrieval task with around 9,500 test images. The fine-tuned model demonstrated performance improvements over the original model.

5 Data Collection

We aimed to build a dataset that prioritizes inclusivity for people rather than focusing solely on languages. Previous efforts have predominantly adopted a language-specific approach, often restricted to officially recognized languages. This has excluded many languages spoken across a linguistically diverse country like India, where hundreds of languages thrive. Additionally, we sought

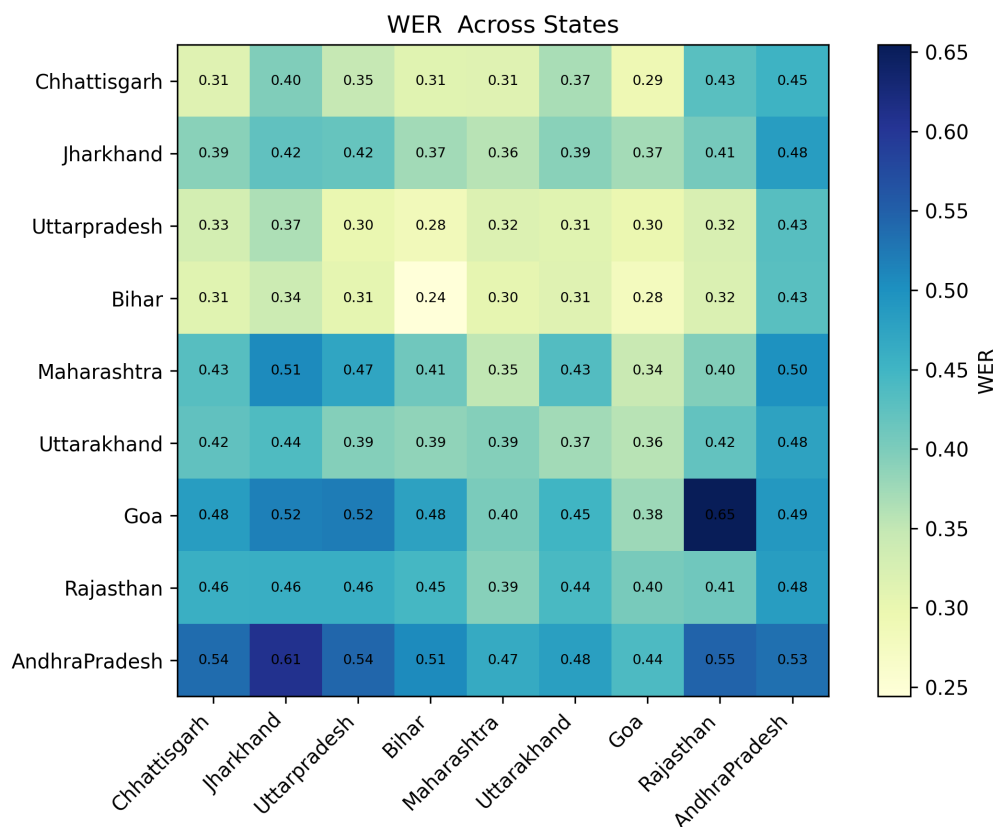
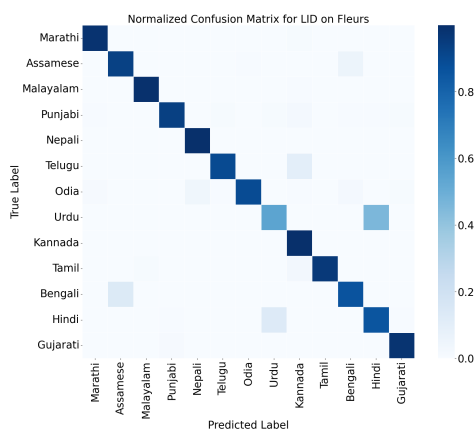
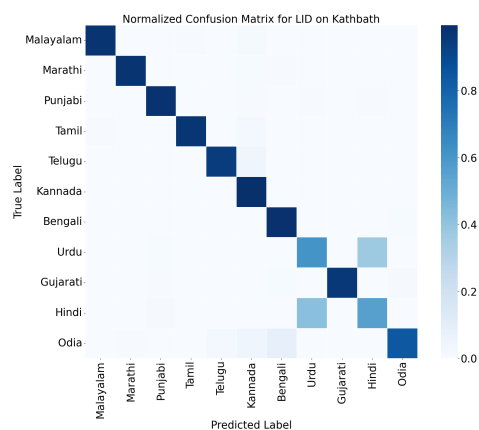


Figure 4: WER heatmap showing state-wise evaluation results for Hindi models fine-tuned on data from individual states. The results highlight that models tend to perform better on geographically proximate states compared to distant ones, even within the same language.



(a) FLEURS:Confusion matrices



(b) KATHBATH:Confusion matrices

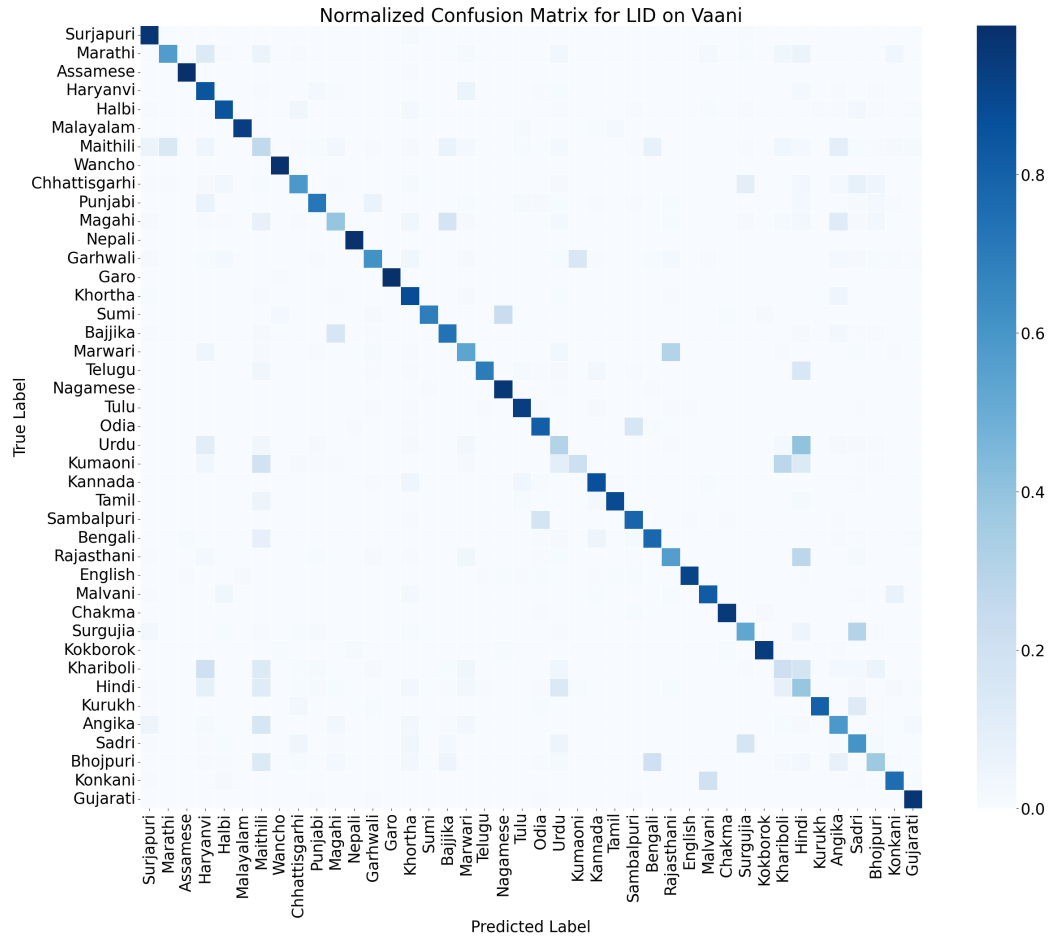


Figure 6: Confusion matrices for LID experiment

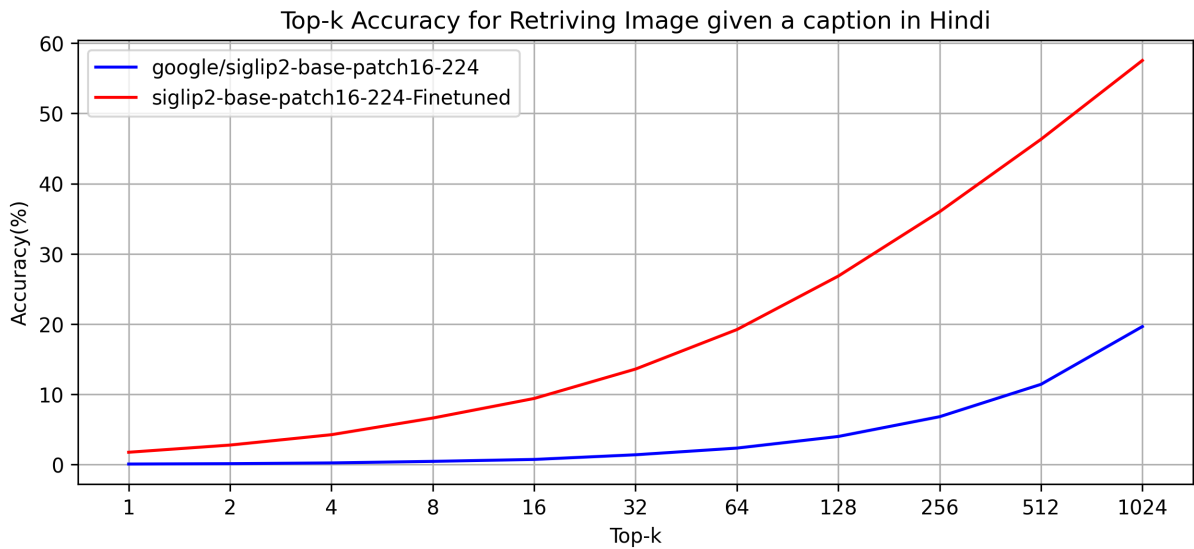


Figure 7: Performance of SigLIP2 fine-tuned on the Hindi portion of the Vaani dataset for the image retrieval task.

to capture more spontaneity in speech, designing our collection process to encourage natural, unbiased expressions, ensuring a more authentic representation of how people communicate.

5.1 Collection Strategy

Our approach to data collection was district-centric rather than language-centric, aiming to capture speech data from individuals across India, representing a wide range of demographics. This included considerations of gender, socioeconomic background, education, and age. To ensure an efficient and scalable process, data collection was organized at the district level. In Phase 1 of the project, we successfully covered 85 districts across 12 states; and in Phase 2 we are covering 80 more districts across 23 states and 3 UTs, ensuring a broad representation of the country’s linguistic and social diversity. This approach allows for a more comprehensive understanding of speech patterns and language use across different communities.

We aimed to capture spontaneous speech from speakers to reflect the natural linguistic variations of their speech. To achieve this, we used carefully selected and specifically prepared images as stimuli. When presented with an image, the speaker was encouraged to describe it, fostering a more natural and creative response. This approach minimized bias and reduced potential inhibitions, which might arise from more structured prompts, such as text-based stimuli. By using images, we provided speakers with greater freedom of expression, leading to more authentic and varied speech compared to the more constrained responses generated by text prompts. Furthermore, compared to conversational data collection, which can be influenced by social factors or biases, image-based prompts allowed for a more genuine and uninhibited language expression from the speaker.



Figure 8: Sample images that used for capturing data for a district

5.2 Collection goals

The districts were selected based on linguistic diversity, with the goal of covering 80 districts in Phase 1 and 85 districts in Phase 2, across 25 states in India. From each district, our goal was to open source around 200 hours of audio data, with 10% of it manually transcribed. Comprehensive metadata was collected for each speaker, including age, gender, languages spoken, education, socioeconomic background, location, and duration of stay in that location. To ensure ethical data use, each participant provided a signed consent form, transferring ownership of the data and enabling its open-source distribution without restrictions. .

5.3 Collection Process

The collection process is operationally challenging, resource intensive, and time-consuming. To address this, we enlisted the support of vendors to manage the process effectively. The vendors established an extensive network of coordinators in each district to facilitate data collection. For audio collection and transcription, we collaborated with three vendors, Shaip(Shaip, 2025), Megdap(Megdap, 2025), and Karya(Karya, 2025) each tasked with providing high-quality audio data, and transcriptions of 10% of those audios, that pass the quality evaluations. We also engaged GTS(GTS.AI, 2025) as the vendor to collect approximately 1500 images from each district.

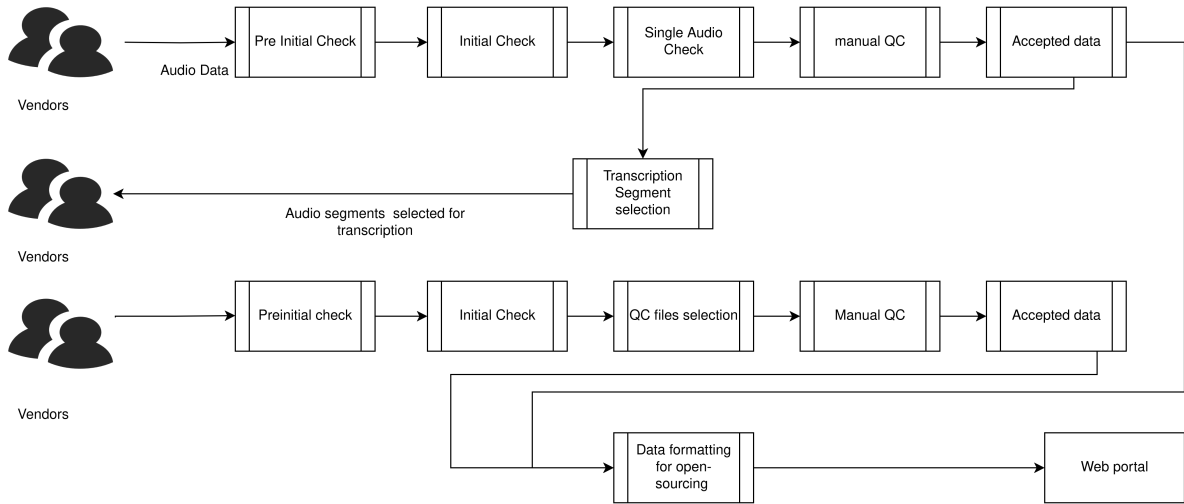


Figure 9: Automated QC-Process flow

The overall process consists of image capturing, speech data collection, and transcription.

5.3.1 Image capturing

In our approach, we presented carefully selected images to participants and asked them to describe the content. For this, we gathered 1,700 to 2,000 images per district, consisting of both district-specific and general images. These images were newly captured for this project with assistance from external vendors. The topics for each district were identified based on local context and shared with the vendors. They were instructed to deliver images in .jpg format, with a resolution of 640x400 pixels and a file size of under 500 KB.

The image collection process adhered to a set of strict specifications to ensure quality, relevance, and ethical standards. All images were required to be physically captured and not sourced from online or third-party media. Each image needed to be clear, well-focused, and free from any blurring or visual distortions. The object relevant to each category had to be prominently visible without any obstructions. Furthermore, images were required to be unique for every district and category, with no duplication permitted across different regions or categories. To maintain privacy and compliance, images were not allowed to include personally identifiable information such as people, rec-

ognizable logos, or privately owned objects. Importantly, the images needed to depict content that could be meaningfully described, as they were intended to prompt spoken responses from participants. Finally, the date of image capture was mandated to be no earlier than July 1, 2023, ensuring the recency and relevance of the content.

To ensure quality, we developed an in-house pipeline for assessing the images delivered by vendors. This process included both automated validation and a human-in-the-loop review. Each image underwent a series of automated and manual checks to verify quality, uniqueness, and compliance with provided guidelines. Only those images that satisfied all the defined criteria were accepted and subsequently used for data collection.

5.3.2 Speech data collection

To facilitate high-quality data collection, we established comprehensive audio and transcription specifications that were shared with the vendors involved in the project. These specifications defined both the technical format of the recordings and the style of data collection to ensure consistency, clarity, and linguistic diversity.

The audio recordings were required to adhere to the following technical standards: a sampling rate of 16kHz, 16 bits per sample, and a single audio channel. Additionally, the audio must be raw —

free from any transcoding or post-processing—to preserve its authenticity and suitability for model training.

Vendors were also given clear instructions regarding the style of data collection. The speech had to be spontaneous, with utterances spoken in response to visual prompts. Participants were shown images selected randomly from a curated pool, and for each image, they were expected to provide 10–20 seconds of effective speech. Instructions emphasized the importance of variety in the spoken responses to ensure linguistic richness and diversity.

To minimize the impact of background noise and maintain recording quality, specific environmental guidelines were also issued. Recordings had to be conducted in quiet environments, free from household noises (e.g., television, kitchen sounds, or water flow) and outdoor disturbances (e.g., birds or traffic). Indoor recordings were to be made in rooms without echo, and speakers were instructed not to record while moving. When using smartphones, all notifications and vibration modes were to be disabled. Furthermore, recordings had to be clean and free of distortions such as clipping, DC drift, signal dropouts, or non-speech interference.

Microphone arrangement was also addressed to ensure optimal audio capture. The recording device had to be placed no more than two feet from the speaker’s mouth, and the distance and orientation were to remain consistent throughout the session. Ideally, the microphone should be positioned directly in front of the speaker, at a minimum distance of one foot, and kept at a fixed location for the duration of recording.

Speaker selection followed strict criteria to ensure nativity, demographic diversity, and balanced representation. Participants were required to be between the ages of 20 and 70, with uniform distribution across this age range. Each district had to include at least 800 speakers, with gender balance maintained. No individual speaker was permitted

to contribute more than 15 minutes of effective speech. All speakers were required to be native residents of the pincode recorded in the metadata, verified using identity documents such as PAN or Aadhaar cards. Participants were encouraged to speak in the language or dialect they typically use at home with family members. Additionally, for each speaker, information about their residential history — specifically where they had lived and for how long since birth — was collected and submitted alongside the recordings.

To collect data from speakers, the vendors utilized their own mobile applications along with a web application for quality control. Coordinators, selected and onboarded from each district, traveled to assigned locations within their districts. They onboarded speakers while ensuring diversity across gender, age groups, education levels, and socio-economic backgrounds. Speakers were registered on the platform with the assistance of coordinators. Initially, they were shown sample images and asked to speak about them. The sample data was then validated by the vendor’s quality control teams. If the data met quality standards, the speaker was officially onboarded to the platform. After onboarding, speakers were presented with a set of images, including district-specific and generic ones, and were asked to record audio. Each speaker contributed up to 15 minutes of audio data. During onboarding, details such as age, gender, known languages, and socio-economic background were also collected. Once the recordings were completed, the application provided an option to submit the data, which was then processed further.

The data collected from users undergoes an initial quality check by the vendors to ensure its integrity. This includes verifying the audio quality, as well as the completeness and accuracy of the metadata. To ensure geographical coverage, pincode information is captured during user onboarding, and the GPS location of the mobile device is used for validation. Once the data is collected and

validated by the vendors, it is submitted for further processing.

5.3.3 Transcription Process

Segments for transcription are selected from the audio data based on a pre-defined criteria and sent back to the vendors. The criteria ensures that a fixed percentage of each speaker's data is transcribed while maintaining diversity across speaker metadata and language. To maintain linguistic and contextual accuracy, specifications require transcribers to be from the same district where the audio data was collected. Vendors recruited and trained transcribers, post which they transcribe the selected audio segments on the basis of guidelines and specifications. Once transcription is completed and undergoes quality control, the transcribed data is submitted for further processing.

6 Quality control

The dataset underwent a comprehensive quality assurance process to ensure its reliability and accuracy across all components, including audio recordings, metadata, transcriptions, and data formatting. The quality evaluation pipeline was meticulously designed to identify and resolve any issues that could compromise the dataset's integrity.

The process incorporated both automated and manual checks. Automated checks evaluated the quality of audio, metadata, and transcriptions. For audio, these checks detected issues such as excessive noise, interruptions, or distortions. Metadata quality was assessed to ensure that all required fields were accurately filled, consistent, and adhered to predefined standards. Similarly, transcription quality was evaluated for accuracy and alignment with the audio content. Data formatting was also reviewed to prevent structural errors that could impact downstream applications. Any inconsistencies, such as mismatched labels, missing data, or errors in transcription, were flagged for correction.

Manual checks were conducted by trained evaluators from the corresponding districts to address finer nuances that automated tools might overlook. These included evaluating subtle audio issues, ensuring metadata accuracy, and verifying the context and correctness of transcriptions. Local evaluators played a crucial role in maintaining contextual and linguistic integrity.

The vendors submitted the data in organized batches through repositories hosted on Google Cloud Platform (GCP). This secure and efficient delivery system ensured that the data transfer process was streamlined. Each batch underwent a final review to confirm compliance with quality standards before being integrated into the larger dataset for further analysis and processing. The various process involved the QC process explained below

6.1 Automated Quality control

In the automated pipeline, audio quality and transcription quality are evaluated separately.

6.1.1 Audio

In this process, the quality of the audio, file format, audio format, and metadata are evaluated at multiple stages based on the complexity of the workflow. This approach enables timely feedback to the vendors, allowing them to take corrective actions and resubmit the data as needed.

Stage 1 - Pre Initial Check

The workflow performs a detailed quality check of the data to ensure its accuracy and consistency across multiple aspects. It validates the structure and content of the metadata and audio files, focusing on several key areas:

Structural Validation: The workflow checks that metadata files follow a strict format. Each required field, such as Speaker ID, Phone Brand, Phone Model, Gender, Age, and more, is evaluated to ensure it is present and correctly formatted. For instance, the fields must contain only allowed characters and end with a newline. Missing

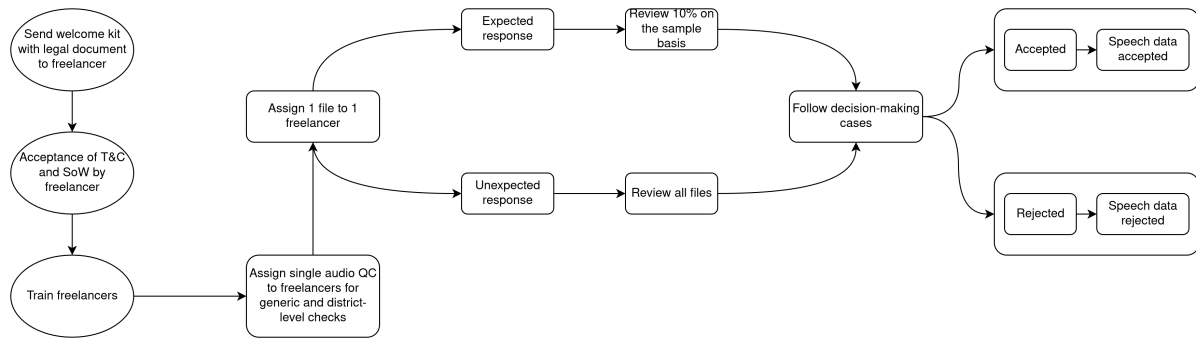


Figure 10: Manual audio QC process

or improperly formatted fields are flagged for correction.

File Naming and Path Validation: The system ensures that file names and paths are consistent and meet predefined standards. Files with spaces, an incorrect number of underscores, or mismatched district or state names are identified as errors. This step also verifies that filenames correctly correspond to the associated metadata and image references.

Data Consistency Checks: The workflow cross-checks metadata with actual data entries. It identifies mismatched or missing speaker IDs, repeated speaker IDs or utterance IDs, and ensures that references to images or other files match those in the database. Any mismatch between metadata and the submitted files is flagged.

Audio Quality and Format Validation: The audio files are checked for proper file format and extension. Files with unsupported formats or incorrect extensions are flagged. The workflow also validates that the total duration of the audio files falls within an acceptable range.

Consent and Supporting Documents: The presence of mandatory consent forms and metadata files is verified. Missing files, such as speaker metadata or consent PDFs, are flagged as critical issues.

Encoding and Formatting Standards: The workflow ensures text files end with a newline, do not contain non-standard Unicode characters, and maintain consistent formatting. This prevents po-

tential issues during data processing.

Geographical and Contextual Validation:

Metadata fields like district and state are cross-verified to ensure they match the expected values. Images or audio files associated with specific districts are checked for consistency with the metadata.

These rigorous checks are designed to catch inconsistencies, formatting errors, and missing data, enabling vendors to receive timely feedback for corrections. This ensures the final dataset is high-quality and ready for downstream processing and analysis.

Stage 2 - Initial Check

The workflow conducts a comprehensive series of checks on both audio data and metadata to ensure the dataset's quality and integrity. These checks validate various aspects of the data and identify inconsistencies or errors that need to be addressed.

Checks performed on the audio files:

Basic Properties: These checks ensure that all audio files meet the required technical specifications: a sample rate of 16 kHz, a single audio channel (mono), and 16-bit precision. Files that fail to meet these standards are automatically flagged for review.

File Integrity: Audio files are examined for redundancy, such as repeated filenames in the current or previous batches. Symbols in file names are checked to ensure they comply with allowed naming conventions.

Segment Details: Segmentation within the audio files is rigorously validated. Issues such as overlapping segment durations, negative durations, or segments shorter than 0.5 seconds are flagged. Speakers exceeding the allowed duration of 15 minutes are also identified.

Silence Analysis: Silence at the beginning, end, or within audio segments is systematically analyzed to ensure clarity and usability of the recordings. Specifically, segments are flagged if they contain silence exceeding 0.3 seconds at the start or end, as this may indicate delayed speech or improper trimming. Additionally, mid-segment silences longer than 1 second are identified, as they can disrupt the natural flow of speech and affect model training or transcription accuracy.

These checks help ensure that the audio quality is suitable for processing.

Consistency: The duration values listed in the `.tsv` files provided by the vendor are compared against the actual audio durations. Discrepancies — such as metadata durations exceeding actual audio lengths — are flagged.

File Presence: Missing or empty `.tsv` files corresponding to `.wav` files are identified. Similarly, cases where image references in metadata do not match the available audio files are flagged.

Metadata Checks:

Demographic Validation: Specific checks are applied to ensure the validity of demographic information provided by speakers. The pincode must correspond accurately to the specified state, ensuring geographical consistency. The speaker's age must fall within a valid range of 20 to 70 years, and the number of years they report staying in a location cannot exceed their age, maintaining logical consistency. Additionally, the language in which the recording is made must be among the languages the speaker has reported as known, ensuring linguistic alignment between the metadata and the audio content.

Consistency Across Batches: Metadata is compared across batches to identify inconsisten-

cies — such as differing demographic details for the same speaker or mismatches between speaker records and batch information.

Speaker Validation: Metadata is analyzed to detect speakers with incomplete or inconsistent data. Speakers with fewer than 10 valid audio files, or those whose metadata conflicts with prior submissions, are flagged for exclusion.

Comprehensive Cross-Validation:

Alignment Between Metadata and Audio: The workflow ensures that metadata entries correctly correspond to associated audio files. Mismatches in filenames, segment durations, or demographic data are highlighted for correction.

These detailed checks help maintain high standards of quality and consistency, facilitating accurate downstream processing and analysis.

Stage 3 - Single Audio Check

The Signal-to-Noise Ratio (SNR) of the audio segments is calculated to evaluate the clarity and quality of the recordings. SNR is a critical metric that measures the level of the desired audio signal relative to the background noise. For each segment, the SNR is compared against a predefined threshold, ensuring that only recordings with an acceptable level of clarity are retained. If any segment fails to meet this threshold, it is flagged for manual review.

6.1.2 Sample Selection for Audio Manual QC process

The files flagged during the single audio check are selected for manual review. Additionally, we ensure that at least one segment from each speaker is included. These segments are randomly sampled so that 10% of the data that passed both the pre-initial and initial checks also undergoes manual quality control.

6.1.3 Files selections for transcription

The pipeline is designed to streamline the selection of audio segments for transcription at the district level, focusing on metadata utilization, speaker diversity, and balanced language represen-

tation. By incorporating statistical analysis, random sampling, and transcript-based strategies, it ensures fair and comprehensive audio file selection.

Segment selection follows systematic approaches that prioritize linguistic diversity and equitable speaker representation while avoiding over-representation of dominant demographics. Transcript-based selection and threshold checks further enhance balance across multiple languages and districts. The iterative process dynamically adjusts to achieve target duration thresholds without compromising diversity.

District-level analysis identifies data coverage gaps through calculated selection rates, ensuring uniform data collection and avoiding the neglect of underrepresented regions. Metadata management is a key aspect, with selected files recorded alongside attributes such as filename, district, language, duration, and selection criteria. This structured approach supports seamless downstream processing and analysis.

The workflow includes robust output and logging systems that save detailed logs, metadata, and summary statistics in specified directories. These outputs offer a transparent view of selection progress and completion, enabling efficient data tracking and ongoing refinement.

6.1.4 Automated Transcription Quality control

Stage 1 - Pre Initial Check

The transcription data undergoes rigorous quality checks to ensure accuracy and consistency. Each segment and its associated metadata are scrutinized for potential errors. The checks validate that segment names and transcriber IDs are correctly assigned, and the segment duration aligns with the number of words, flagging segments where the word count falls below a defined threshold. Transcriptions are evaluated for the presence of invalid elements such as numbers, incomplete sentences, or missing punctuation like open or un-

closed brackets. Additionally, checks ensure that non-native characters are not used, and any missing or mismatched segments between the transcription files and the original audio data are identified.

The transcription metadata undergoes a parallel validation process. Critical fields like Transcriber ID, Gender, Socio economic, Education, Pincode, and Experience years are verified for presence and correctness. Errors such as duplicate fields, invalid filenames, or missing entries are flagged. Metadata validations extend to logical consistency checks, such as ensuring the transcriber's age and experience values are non-negative and that age is greater than or equal to experience. Invalid symbols in language-related fields and discrepancies in transcriber information, such as mismatched or missing demographic details, are also highlighted. Finally, checks confirm the proper formatting of metadata files, ensuring no blank lines, incorrect colon usage, or repeated entries within a batch. These thorough validations maintain the reliability of transcription data for downstream use.

Stage 2 - Initial Check

The transcription data undergoes advanced quality checks to ensure linguistic and structural consistency, leveraging both statistical and logical evaluations. Each segment is analyzed for adherence to defined thresholds, such as LM loglikelihood, where transcriptions must exhibit linguistic plausibility within acceptable bounds. The Word Error Rate (WER) is also assessed, with segments exceeding predefined thresholds flagged for review. Scripts used in the transcription are verified for uniformity, ensuring that characters from unintended or mixed scripts are not present. A critical ratio analysis evaluates the relationship between the number of words and segment duration, flagging segments that are either too verbose or too sparse relative to their length. Additionally, the presence of repeated words is measured, and segments with a high repetition ratio are identified.

Language consistency is scrutinized by cross-

checking the transcription language with the expected language for the segment. A mismatch triggers an error for correction. The geographic relevance of the transcription is also validated, comparing pincode information against the expected region. A significant difference between the pincode in the transcription metadata and the designated area triggers a warning. These checks collectively ensure that the transcriptions are linguistically accurate, structurally coherent, and contextually relevant, contributing to a high-quality dataset for downstream applications.

6.2 Manual Quality control

To ensure effective quality control, we have assembled and trained a team of validators from across the districts, supervised by experienced data annotation professionals from across hailing from different native languages. Data validation is carried out at the district level by validators from each respective district. A multi-level quality check process is in place to guarantee the highest data quality. The first level of quality checks is conducted by the experts, with random samples being reviewed by a supervisor. If quality issues are identified greater than a set threshold (10 percent), the entire dataset is sent for further validation by additional experts to ensure accuracy and consistency.

6.2.1 Audio

In this manual quality check for audio, several important aspects are assessed to ensure clarity and relevance. First, it is confirmed that humans are speaking in the audio and that only one person is speaking about the image. The speaker's gender is identified, and the language expert affirms that the audio is understandable. Additionally, the speaker is identified as someone from the district, ensuring contextual accuracy. The audio is free from disturbances that could hinder comprehension, and it sounds like a complete, coherent sentence. Furthermore, it is verified that the audio content is

relevant to the image, and no flags were raised on the speaker's appropriateness. Privacy and confidentiality are also prioritized by ensuring there is no personally identifiable information (PII) in the image or audio files. This process helps maintain high standards of transcription and ensures the quality and relevance of the audio content.

6.2.2 Transcription

To ensure the quality of speech transcription, a thorough manual check is conducted with the help of a language expert. This process involves verifying that the transcribed text exactly matches the audio, ensuring the script aligns with the language being spoken. The language expert also checks for repetition of words, confirming that the transcription remains fluent and without unnecessary redundancy. Additionally, the nature of the speech is assessed to confirm whether it is natural, as opposed to unnatural, read, or guided speech, ensuring authenticity in transcription. Privacy and confidentiality are also prioritized by ensuring there is no personally identifiable information (PII) in the image or audio files. These steps, supported by the language expert's insights, contribute to the accuracy and reliability of the transcription process.

7 Conclusions and Future work

We present a multimodal dataset comprising 31,270 hours of audio and 2,067 hours of transcriptions, collected using approximately 289K images. The dataset is accompanied by rich metadata and has undergone a thorough curation process. Our limited experiments demonstrate the utility of the data across different modalities. We plan to expand data collection to additional districts and release the dataset to support a more inclusive Digital India.

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A Transcription Guidelines

Vaani Audio Transcription Guidelines (Last updated: Oct 3, 2022)

The following transcription guidelines are adapted from the Digital India Bhashini Project (Sep 12, 2022 version), which in turn are inspired by similar guidelines created by a commercial transcription agency and by NIST. General Principles Transcribe a word only if you can hear and understand it properly. If the spoken word/text cannot be understood due to the speaker’s manner of speech then mark it as [unintelligible]. On the other hand, if the spoken text cannot be heard due to poor recording, volume or noise then mark

it as [inaudible]. An audio segment will be considered as invalid if more than 25% of the segment is inaudible. Do not paraphrase the speech. Do not correct grammatical errors made by the speakers. In the case of misspoken words, transcribe the one with a spelling mistake followed by the correct spelling within curly braces. Example: If a speaker pronounces "remuneration" as "renumeration" then it should still be transcribed as "renumeration remuneration". Do not expand spoken short forms (e.g., ain't, don't, can't, it should be retained as it is) Retain colloquial slang as it is (e.g., gotcha, gonna, wanna, etc).

If the spoken words belong to a language that the transcriber does not know then it should be tagged as <UNKNOWN_SEGMENT> with the appropriate timestamps. If transcription of a sentence contains <UNKNOWN_SEGMENT> for more than 25% of the sentence, the entire sentence should be transcribed as <UNKNOWN_SEGMENT>. Every pause more than 0.5sec should be transcribed as <PAUSE>. Verbatim transcription The speech should be transcribed verbatim.

However, the following rules should be used for transcribing errors made by the speakers. Errors that should be transcribed as it is: Speech errors: "I was in my office, no sorry, home" should be transcribed as it is. Slang words: kinda, gonna, wanna, etc should be transcribed as it is. Repetitions: "I have I have got the book" should be transcribed as it is.

Errors that should be transcribed: False starts: "I, um, er, I was going to the mall" should be transcribed as it is. Filler sounds: um, uh, er, hmm, etc. should be transcribed as it is. Stutters: "I w-w-went t-t-to the mall" should be transcribed as it is. Following the guidelines for the SWITCHBOARD corpus there should be a hyphen between the stutters as shown in the above sentence.

Non-speech (acoustic) events

Foreground sounds made by the speaker should be transcribed. These include lip smacks, tongue clicks, inhalation and exhalation between words, yawning coughing, throat clearing, sneezing, laughing, chuckling, etc. The categories speci-

fied in Table 1 in the guidelines for the SWITCHBOARD corpus, should be used. Names, titles, acronyms, punctuations, and numbers Proper names should be transcribed in a case-sensitive manner in applicable languages (e.g., English). Initials should be in capital letters with no period following. For example: "M K Stalin would be sworn in as the Chief Minister". Titles and abbreviations are transcribed as words. For example: Dr. → Doctor except if the abbreviated form is pronounced as it is. For example, if the speaker says "Apple Inc" (instead of "Apple Incorporate"), the word 'Inc' should be transcribed. Punctuation marks should be used in transcription as appropriate. For example: "don't" Apart from the letters (or Aksharas) only the following special characters can be used - COMMA (,), FULLSTOP (.), QUESTION MARK (?), APOSTROPHE ('). UNDERSCORE (' _ ') to be used only for acronym (see the next bullet). Purna viram (l) may be used to indicate end of a sentence only.

Numbers should be transcribed as they are spoken and normalized to word form (no numerals). For example: 16 → sixteen, 112 → one hundred and twelve. Times of the day and dates: always capitalize AM and PM (wherever applicable). When using o'clock, spell out the numbers: eleven o'clock.

Incomplete utterances These are utterances which are incomplete because the speaker forgot what he wanted to say or was stopped mid-way and corrected an error or was interrupted by someone. Indicate such utterances by putting a '–' at the end of the utterance as opposed to a full-stop or question mark.

Table 2: f1-scores (%) of our whisper-medium + pooled attention + ANN model across languages on our test set, and two external test sets: FLEURS, and Kathbath.

Language	Vaani Test	FLEURS	Kathbath
Angika	54.4	—	—
Assamese	98.5	90.8	—
Bajjika	71.3	—	—
Bengali	73.6	88.1	88.6
Bhojpuri	44.2	—	—
Chakma	95.6	—	—
Chhattisgarhi	66.5	—	—
English	93.8	—	—
Garhwali	67.9	—	—
Garo	99.2	—	—
Gujarati	93.8	80.3	78.0
Halbi	82.4	—	—
Haryanvi	68.8	—	—
Hindi	25.9	40.0	39.3
Kannada	84.4	93.9	91.1
Khariboli	25.4	—	—
Khortha	78.5	—	—
Kokborok	94.2	—	—
Konkani	76.0	—	—
Kumaoni	28.2	—	—
Kurukh	88.3	—	—
Magahi	46.2	—	—
Maithili	21.0	—	—
Malayalam	95.6	99.2	98.2
Malvani	81.6	—	—
Marathi	65.0	63.2	61.1
Marwari	55.3	—	—
Nagamese	81.2	—	—
Nepali	96.8	88.7	—
Odia	76.9	74.3	47.0
Punjabi	77.7	87.7	85.2
Rajasthani	61.0	—	—
Sadri	51.1	—	—
Sambalpuri	77.4	—	—
Sumi	81.7	—	—
Surgujia	58.1	—	—
Surjapuri	87.7	—	—
Tamil	92.1	97.0	96.9
Telugu	80.8	92.9	95.5
Tulu	90.3	—	—
Urdu	34.4	33.2	51.0
Wancho	96.5	—	—
Overall Accuracy	74.8	76.1	65.8